EXPERIMENT 9

BI Security: Authentication, Authorization and Access Control using Microsoft Power BI

AIM

To understand and implement Business Intelligence security concepts including authentication, authorization, role-based access control, and data protection mechanisms using Microsoft Power BI.

TOOLS / SOFTWARE REQUIRED

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Microsoft Excel

DATASET AND POWER BI FILE USED

Global_Superstore_2018.xlsx loaded into Exp_9_BI.pbix. The model is the star schema of earlier experiments: Fact_Sales (centre) with Dim_Customer, Dim_Product, Dim_Region and Dim_Date, plus a Business Scorecard table. The report pages are Page 1, Relational Report, Matrix Report, Comparative Analysis, Executive Dashboard and Business Scorecard. Dataset size: 51,290 order lines, total sales $12,642,502, total profit $1,467,457, 23 regions in 5 markets.

THEORY

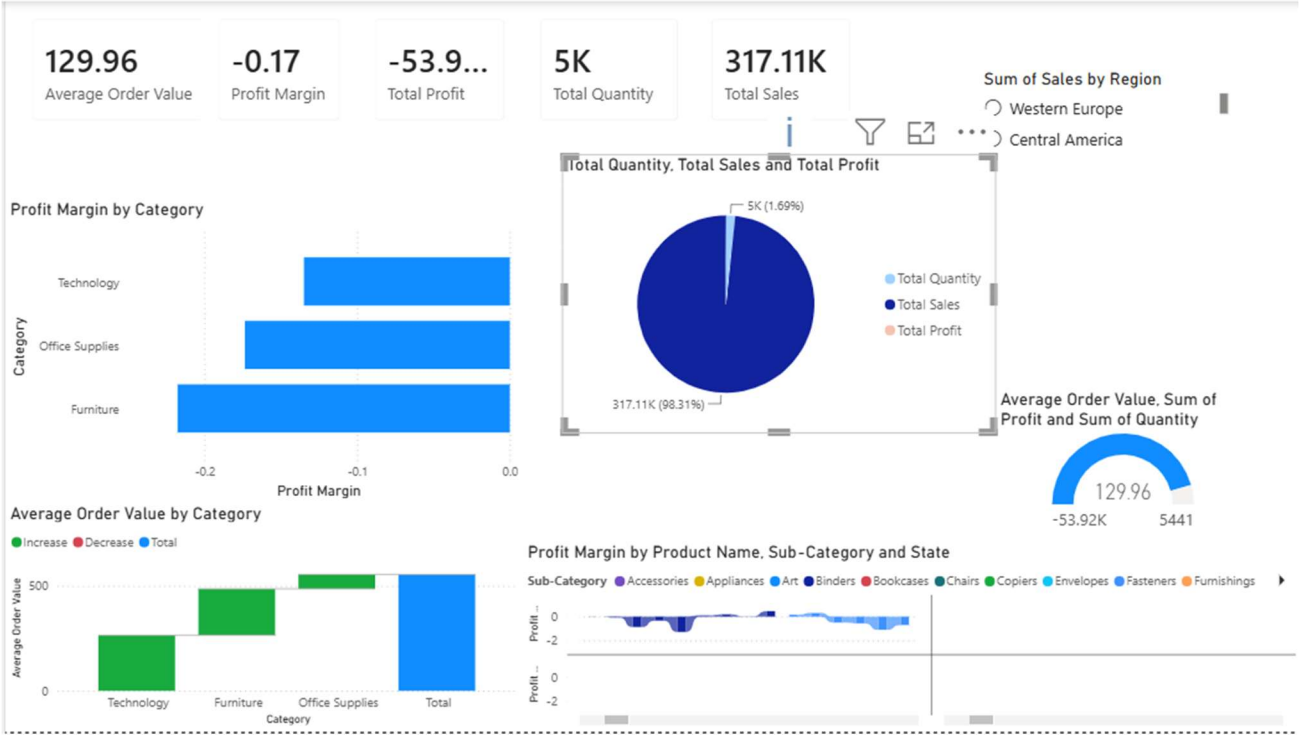
Security is a critical component of Business Intelligence systems. BI platforms hold sensitive organizational information that must be protected from unauthorized access. Authentication verifies the identity of a user, while authorization determines the resources and actions that the user may use. Access control ensures that users see only the data relevant to their responsibilities. Power BI supports security through workspace roles, sharing and app permissions, Row-Level Security (RLS) and role-based access control.

BUSINESS SCENARIO

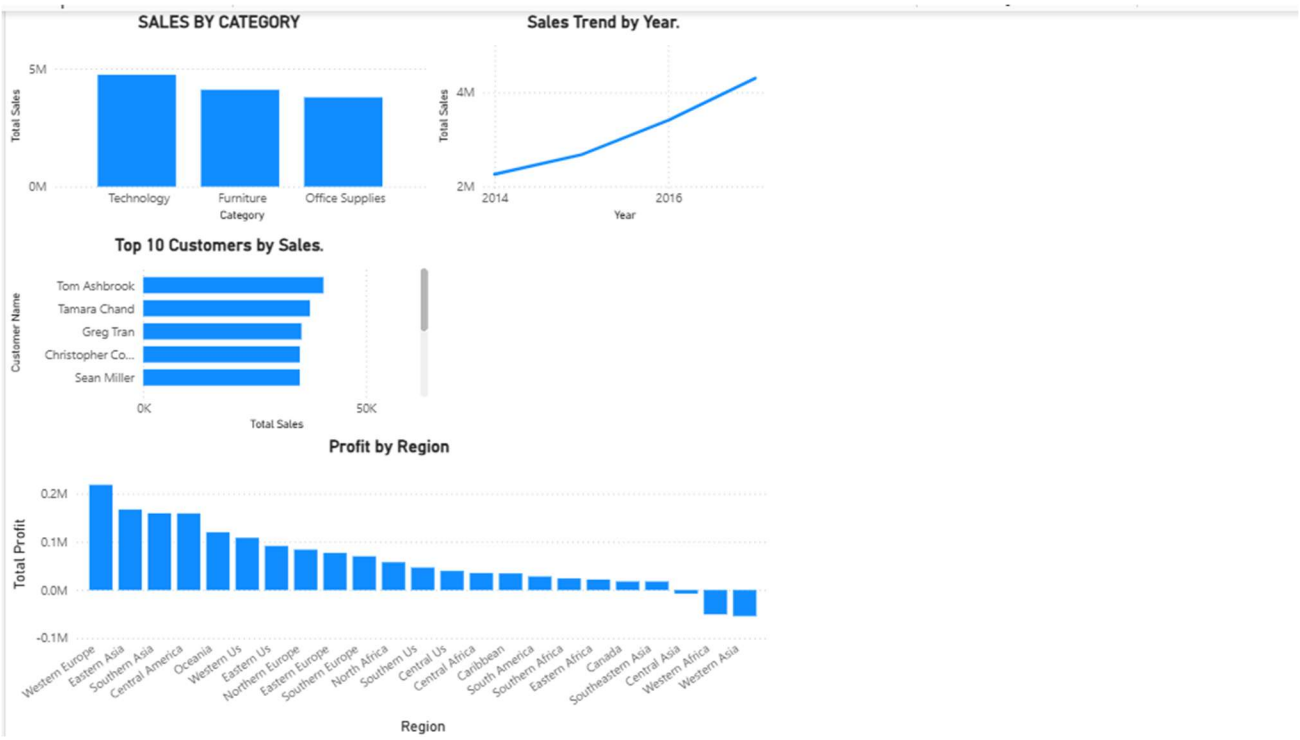
Global Superstore has multiple departments including Sales, Finance and Regional Management. Each department requires access only to information relevant to its responsibilities. As BI Security Administrator, a secure access strategy is designed for the organization's reports and dashboards.

CONDUCT - PART A: USER AND ROLE IDENTIFICATION

Step 1-2: Users and their information requirements



Step 3: Role Matrix (reports each role can access)



CONDUCT - PART B: AUTHENTICATION ANALYSIS

Step 4-5: Power BI uses Microsoft Entra ID (Azure Active Directory) to authenticate every user before any report, dataset or workspace can be opened. The three methods are compared

below.

Method	How it works	Advantages	Weaknesses	Suitability
Username and password (local)	User enters a stored	Simple to set up	Can be guessed, phished, reused;	Not recommended



Step 6: Advantages of secure authentication

- Only verified users can reach confidential sales and profit data.
- Every access is traceable to a person through audit logs.
- Leavers lose access at once because the account is disabled in one place.
- Conditional access can block risky sign-ins (unknown location or device).
- MFA keeps data safe even when a password is stolen.

CONDUCT - PART C: AUTHORIZATION AND ACCESS CONTROL DESIGN

Step 7-9: Access Control Matrix. Permissions: View (read reports), Edit (change report/dataset), Share (give access to others), Admin (manage workspace, security, settings). Workspace roles used: Viewer, Contributor, Member, Admin.

Step 10: Justification of permissions

- Least privilege: every user receives only the permission needed for the job.
- Executives only view; they do not need to modify datasets, so accidental changes are avoided.

- Finance can edit its own report because profitability analysis changes frequently; it cannot administer the workspace.
- Regional Managers view through a published App and RLS, so they cannot download or see other regions.
- Only the BI Administrator manages security roles and workspace access (separation of duties).

CONDUCT - PART D: ROW-LEVEL SECURITY (RLS) PLANNING

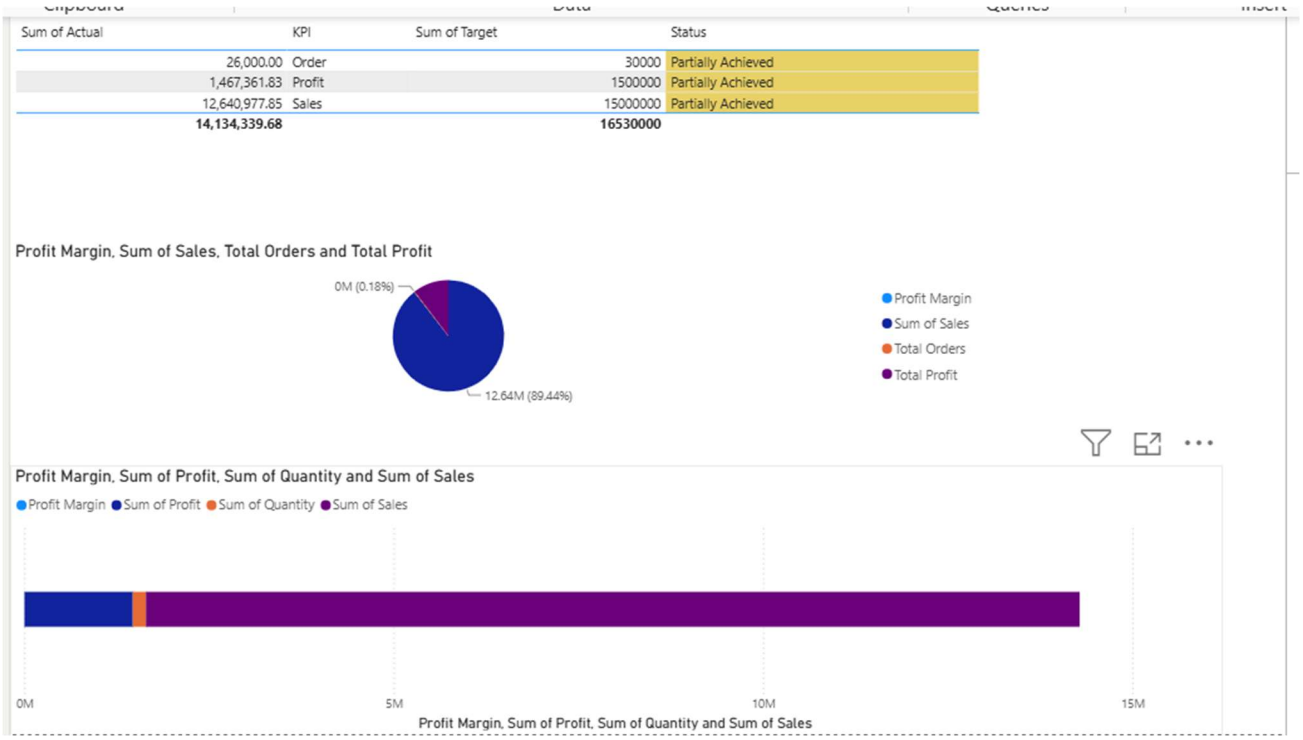
Step 11: RLS strategy. The dataset has a relationship Dim_Region -> Fact_Sales, so filtering Dim_Region[Region] automatically filters all sales rows. Static roles are created for fixed users and dynamic RLS is used when many managers exist.

Static RLS roles (Power BI Desktop > Modeling > Manage roles):

- Role 'Regional Manager - Eastern US' Table Dim_Region DAX filter: [Region] = "Eastern US"
- Role 'Regional Manager - Western Europe' DAX filter: [Region] = "Western Europe"
- Role 'Regional Manager - Southern Asia' DAX filter: [Region] = "Southern Asia"
- Roles 'CEO' and 'Finance Manager': no filter (see all data)

Dynamic RLS (recommended): create a table UserSecurity with columns UserEmail and Region (one row per manager, based on the People sheet), relate it to Dim_Region[Region], and create one role 'Regional Managers' with this filter on UserSecurity:

- [UserEmail] = USERPRINCIPALNAME()



Testing: Modeling > View as > select role (and optionally another user). Then publish and, in Power BI Service, open Dataset > Security and add each user or group to its role. Note: RLS is not applied to workspace Admins, Members and Contributors, so managers must only be given

Viewer access or access through an App.

Regional Managers from the People sheet (24 regions): e.g. Eastern US - Dolores Davis, Western Europe - Gilbert Wolff, Southern Asia - Chandrakant Chaudhri.

RLS effect on the data (from Global Superstore)

Role (RLS filter)	Rows visible	Sales visible	Profit visible	Share of company sales
CEO / Finance (no filter)	51,290	$12,642,502	$1,467,457	100%
Regional Manager - Eastern US (Dolores Davis)	2,848	$678,781	$91,523	5.4%
Regional Manager - Western Europe (Gilbert Wolff)	5,883	$1,731,930	$218,434	13.7%
Regional Manager - Southern Asia (Chandrakant Chaudhri)	2,655	$866,573	$159,336	6.9%

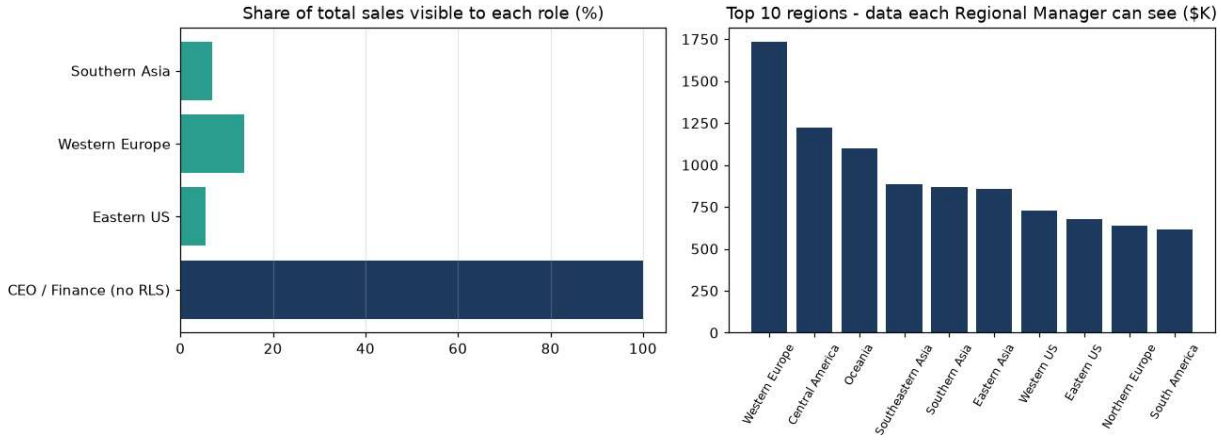


Figure 1: Data visible to each role under RLS (analysis of the Global Superstore data; replace with Power BI 'View as role' screenshot).

Step 12: How RLS improves data security

- Data is filtered at the dataset, not just hidden on a page, so a user cannot see other regions by changing a filter, exporting or using Analyze in Excel.
- One report serves every manager, so there are no separate copies to maintain.
- Dynamic RLS scales: adding a manager only requires a new row in UserSecurity.
- It limits damage if an account is compromised, because only one region is exposed.
- It supports compliance and gives clear accountability for who may see which data.

CONDUCT - PART E: SECURITY RISK ASSESSMENT

Step 13-14: Risks and controls

Risk	Description	Likelihood	Impact	Control / Mitigation
Unauthorized access	Stolen or shared credentials give access to confidential sales and profit data	Medium	High	Entra ID sign-in, MFA, conditional access, no shared accounts
Excessive permissions	Users with Admin/Member roles can bypass RLS and edit datasets	High	High	Least privilege, Viewer role through Apps, quarterly access review
Data leakage	Export to Excel/PDF, public links, 'Publish to web', forwarding of emails	Medium	High	Disable Publish to web, restrict export, sensitivity labels, DLP policies
Weak authentication	Passwords only, no MFA, no lockout	Medium	High	Enforce MFA for all and password policy
RLS misconfiguration	Wrong filter or user missing from a role shows all or no data	Medium	Medium	Test with 'View as', document roles, test after each change
Orphan accounts	Leavers or role-changers keep access	Medium	Medium	Automated deprovisioning, group-based access
No audit trail	Misuse not detected	Low	Medium	Enable audit logs, usage metrics, alerts on unusual activity

SECURITY RECOMMENDATIONS SHEET

No.	Recommendation	Owner	Priority
1	Enforce MFA and organizational (Entra ID) sign-in for all Power BI users	IT Security	High

2	Implement dynamic RLS on Dim_Region using a UserSecurity table	BI Developer	High
3	Give business users Viewer access via Power BI Apps; limit Admin/Member to the BI team	BI Administrator	High
4	Use security groups, not individual users, for roles and workspaces	BI Administrator	Medium
5	Apply sensitivity labels and restrict export and sharing for profit and cost data	Data Governance	High
6	Disable 'Publish to web' and review external sharing	BI Administrator	High
7	Review access and RLS membership quarterly	BI Administrator	Medium
8	Enable audit logs and monitor usage and unusual downloads	IT Security	Medium

DELIVERABLES

- Role and Permission Matrix - Part A and Part C
- Access Control Matrix - Part C
- Row-Level Security Design Document - Part D
- Security Risk Assessment Report - Part E
- Security Recommendations Sheet - Part E

RESULT

A secure BI access strategy was successfully designed using authentication, authorization, and access control principles. User roles, permissions, and security measures were defined to protect organizational data and ensure controlled access.

LEARNING OUTCOMES

- Explained authentication, authorization, and access control concepts in BI systems.
- Identified security requirements for different organizational roles.
- Analyzed access control mechanisms and permission structures.

- Evaluated security risks associated with BI deployments.
- Designed secure access strategies for BI environments.